

Particles Escaping From a Box

Basit Reyaz

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Abstract

This report explores the escape dynamics of diffusing particles from a two-dimensional square cavity of side length L with a centered exit hole of size h ($0 < h < L$). Particles, initialized at random positions, undergo Brownian motion, and the half-time, the time for half the particles to escape, is computed using random-walk simulations. Key parameters such as exit hole size and diffusivity are analyzed. Simulations revealed that larger openings significantly reduce the half-life by increasing the probability of particle escape, while smaller openings act as restrictive bottlenecks, prolonging the escape process. The step size also impacts the dynamics: smaller steps result in slower diffusion and extended escape times, whereas larger step sizes facilitate faster traversal, enhancing the escape rate. Results highlight the significant impact of geometry and diffusion on escape times, offering insights into confined transport phenomena relevant to natural and engineered systems.

1 Introduction

In many physical and biological systems, understanding the escape dynamics of particles from confined geometries is a fundamental problem with applications ranging from molecular diffusion to ecological modeling. A simple yet insightful model to study such behavior is the diffusion of particles within a bounded two-dimensional square cavity with a centered exit hole. This problem provides a computational framework to investigate how geometric constraints, diffusivity, and initial particle distribution affect the escape time dynamics.

The setup consists of a square cavity of side length L , where particles are initialized at uniform random positions within the box. At the center of one of the sides of the square lies an exit hole of size h ($0 < h < L$), through which particles can escape. The primary quantity of interest in this study is the half-time of the cavity, defined as the time required for half of the particles to escape through the exit hole.

The results of this work provide insights into the interplay between geometry and diffusion in determining escape times, and the findings may have broader implications for studying diffusion-driven escape phenomena in both natural and engineered systems.

2 Method

Our objective is to design an efficient algorithm to simulate the Brownian motion of particles within a two-dimensional box of side L . The box features an opening of size d on one of its sides, allowing particles to gradually escape through the opening. The particles undergo Brownian motion inside the box due to random collisions, and over time, a fraction of them exits the box. Using these simulations, we aim to determine the **half-life**, defined as the time it takes for half of the particles to leave the box. In addition, we will systematically study the dependence of the half-life on various parameters, including the size of the opening (d), the step length (ℓ , which is directly related to the diffusion constant D), and the initial number of particles inside the box (N_0). By varying these parameters, we aim to understand how they influence the escape dynamics of the particles. This study provides valuable insights into diffusion processes in confined geometries and has applications in understanding molecular transport through membranes or particle motion in microfluidic devices.[1]

Brownian motion is a continuous-time and continuous-space stochastic process that describes the random motion of particles suspended in a fluid, resulting from collisions with surrounding fluid molecules. In contrast, numerical simulations typically employ discrete time and space steps. Despite this difference, the random walk model, which is a discrete counterpart of Brownian motion, provides an excellent approximation for simulating the behavior of Brownian particles. By carefully adjusting the step size and time interval, random walk simulations can effectively capture the key features of Brownian motion in a computational setting.

Random walk step length d is related to the diffusion constant D in two dimensions through,[2]

$$D = \frac{d^2}{4\Delta t} \tag{1}$$

Where Δt is time taken for a single step, which we can define to be 1 units.

The first step is to simulate the random walks of multiple particles in two-dimensional space. This can be done in two ways: by performing a random walk on a lattice grid of x and y coordinates, or by using angular displacements to determine the direction of each step. The method involving angular displacements is preferred, as it effectively covers a wider range of points, offering a more continuous and realistic representation of the random motion compared to the discrete lattice-based approach.

We simulate these random walks and visualize them in Figure 1. To keep the plot clear and focused, we display the trajectories of only four particles out of the total 1000 particles simulated in our code. Now moving on to a more complex case, we enclose

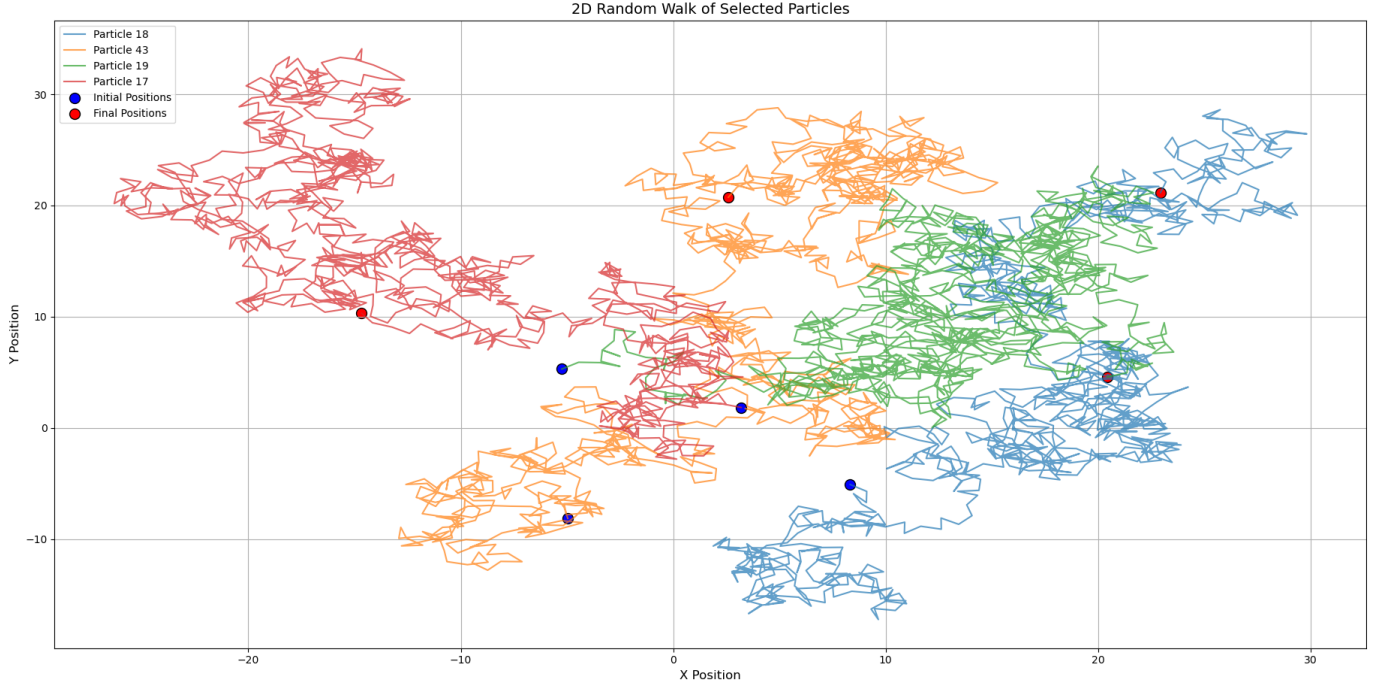


Figure 1: 2D Random Walk

the space by a box of dimensions L . The walls of the box are reflective, the particles bounce back after striking them. The reflective boundary condition ensures that particles remain within a box of side length L , centered at the origin. If a particle reaches or exceeds the boundary during its random walk, its position is reflected back into the box. Consider a particle with an initial position x_i and after a step, its new position is x_f . The reflection condition for the x -direction can be written as:

$$x_f = \frac{L}{2} - (x_f - \frac{L}{2}) \quad \text{for } x_f > \frac{L}{2}$$

which simplifies to:

$$x_f = L - x_f$$

This ensures that if the particle moves beyond the positive boundary $\frac{L}{2}$, it is reflected back by subtracting the excess distance from the boundary point.

Similarly, if the particle moves beyond the negative boundary $-\frac{L}{2}$, the reflection is given by:

$$x_f = -L - x_f \quad \text{for } x_f < -\frac{L}{2}$$

The same equations apply for y coordinate as well.

This reflection scheme ensures that the particle stays within the box, effectively "bouncing" off the boundaries and continuing its random walk inside the domain. The simulation of the random walks inside such box is shown in Figure 2.

Next thing we want to do, is to have an opening of size h centered at one of the sides of the box, through which the particles can escape from the box. The half-life calculation in the algorithm tracks the time it takes for half of the particles to escape through the central opening in the box. During the simulation, each particle's position is updated at each time step, and the escape condition is checked: if a particle's y -position exceeds the top boundary ($L/2$) and its x -position is within the opening (i.e., $|x| \leq d/2$), the particle is considered to have escaped. The algorithm continuously updates which particles have

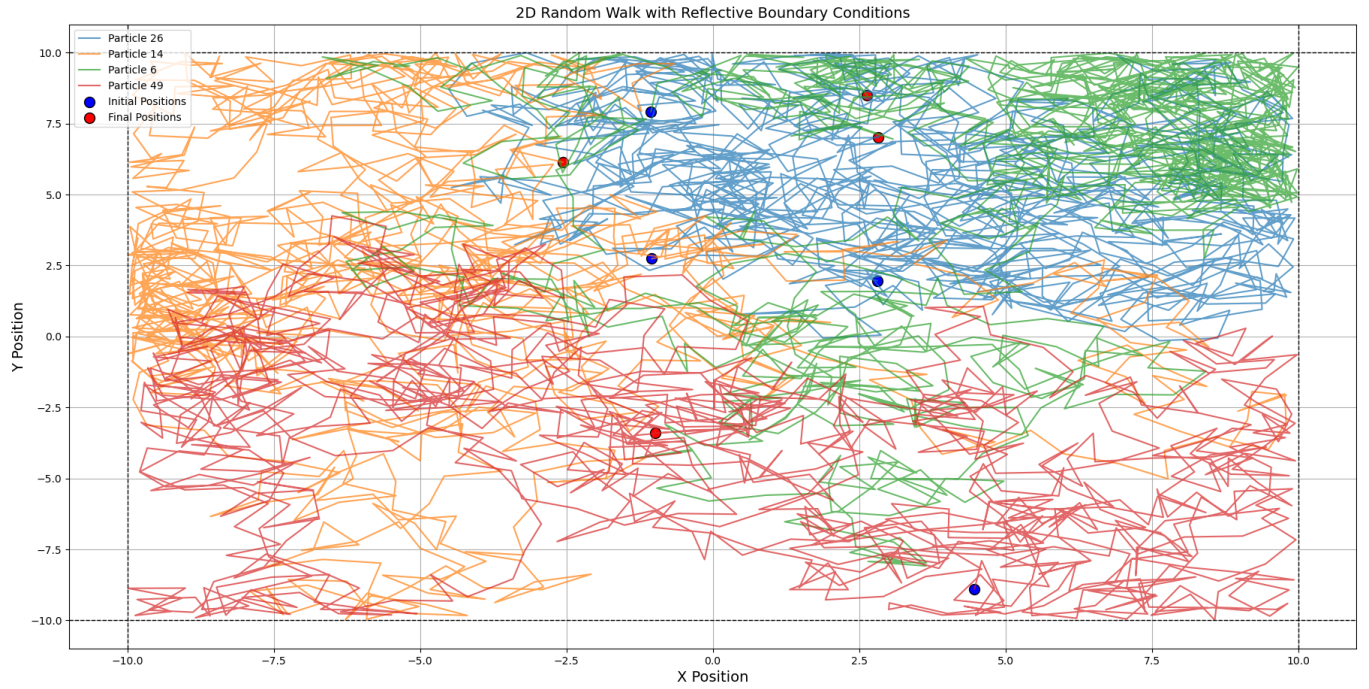


Figure 2: Random walk inside a box of size 20 units.

escaped and stops the simulation once half of the particles have escaped. The half-life is then determined by the number of time steps it takes for this condition to be met. If fewer than half of the particles escape by the end of the simulation, the total number of steps is returned as the half-life.

The half-life itself is not of much interest, rather what we are looking for is its dependence on various physical parameters like the size of the opening and number of particles.

3 Results

After running the simulation, we can vary the size h of the opening through which the particles can escape the box, then observe the behavior of the half-life with the variations in h . The relation between the half life $t_{1/2}$ and the opening size h ,

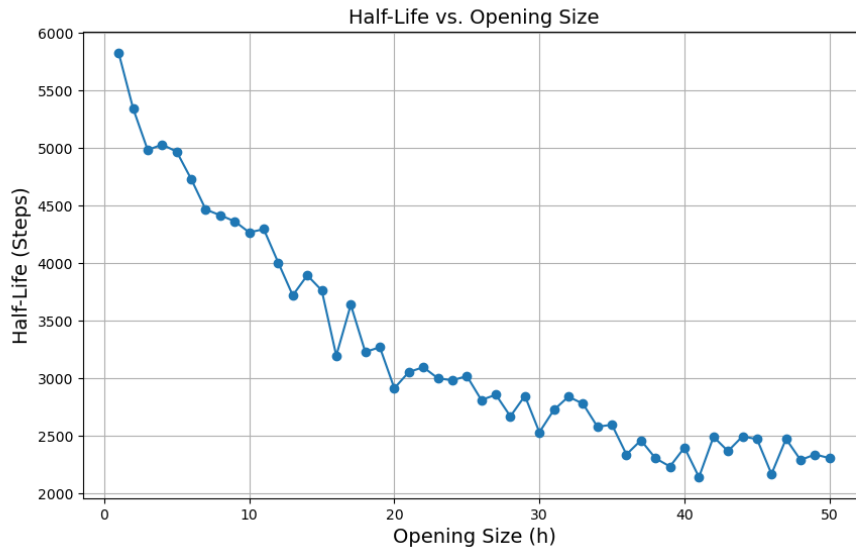


Figure 3: Half-Life in a single simulation

generally follow inverse law,[3],

$$t_{1/2} = \frac{K}{h}$$

Where K is some constant. This behavior can be observed from Figure 3 as well. This behavior is quite intuitive, as h increases the corresponding decrease in half life becomes less prominent since the opening has already become large enough for the particles to get through easily.

There are small peaks and valleys in the plot, these arise from statistical fluctuations, we can smoothen them out by running multiple simulations, then averaging different values of half life for individual sizes of the hole h .

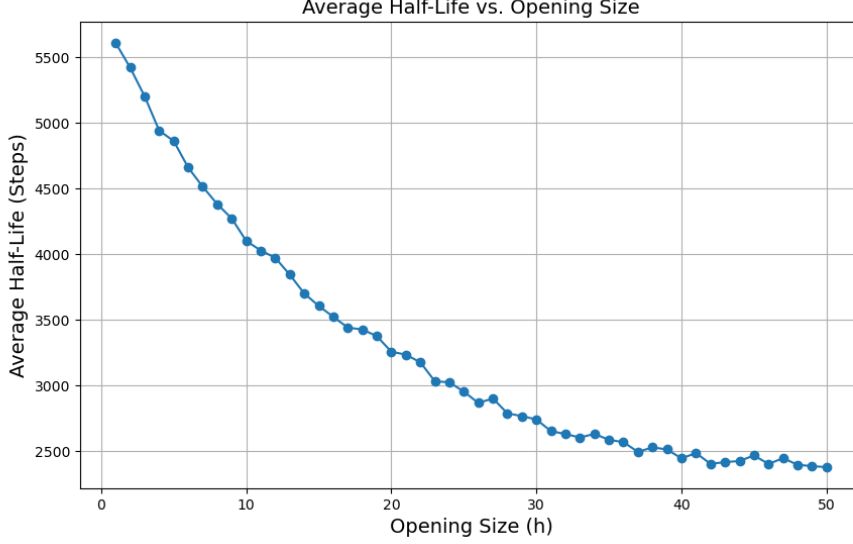


Figure 4: Half-life averaged over multiple simulations

We can also analyze variations in half-life with the step length(d), shown in Figure 5. This plot is also intuitive to understand.

- For small step lengths, the particles may not even leave the box, if the total number of steps is not large enough. Thus half-life will be ver high.
- As the step length increase, the probability of moving towards the opening increases, thus half life decreases significantly.

The simulation of particle escape from a 2D box with a central opening reveals a clear relationship between the step size of the random walk and the calculated half-life of the system. As the step size increases, the particles explore the box more rapidly, leading to a reduction in the time it takes for half of the particles to escape. This behavior aligns with the physical intuition that larger steps allow particles to traverse the domain and encounter the escape region more quickly.

However, the observed trend is nonlinear, reflecting the combined effects of reflective boundary conditions, the finite size of the box, and the constrained opening. For very small step sizes, particles exhibit a more diffusive motion, resulting in slower escape times. On the other hand, excessively large step sizes may lead to inefficient sampling of the domain near the opening, causing deviations from the expected monotonic decrease in half-life.

4 Conclusion

This study investigated the escape dynamics of particles performing 2D random walks within a confined square box, with particular emphasis on the relationship between the half-life—the time required for half of the particles to escape—and two key parameters: the size of the central opening and the step size. The simulations demonstrated that increasing the size of the opening drastically reduces the half-life, as larger openings provide a significantly higher probability for particles to exit the box. Conversely, smaller openings act as restrictive bottlenecks, leading to prolonged escape times due to the limited area available for particle egress. The step size also played a crucial role, with smaller steps resulting in slower diffusion and longer escape times, while moderate step sizes enhanced the efficiency of escape by enabling particles to traverse the box more effectively. These results illustrate the delicate balance between system geometry and particle motion, offering

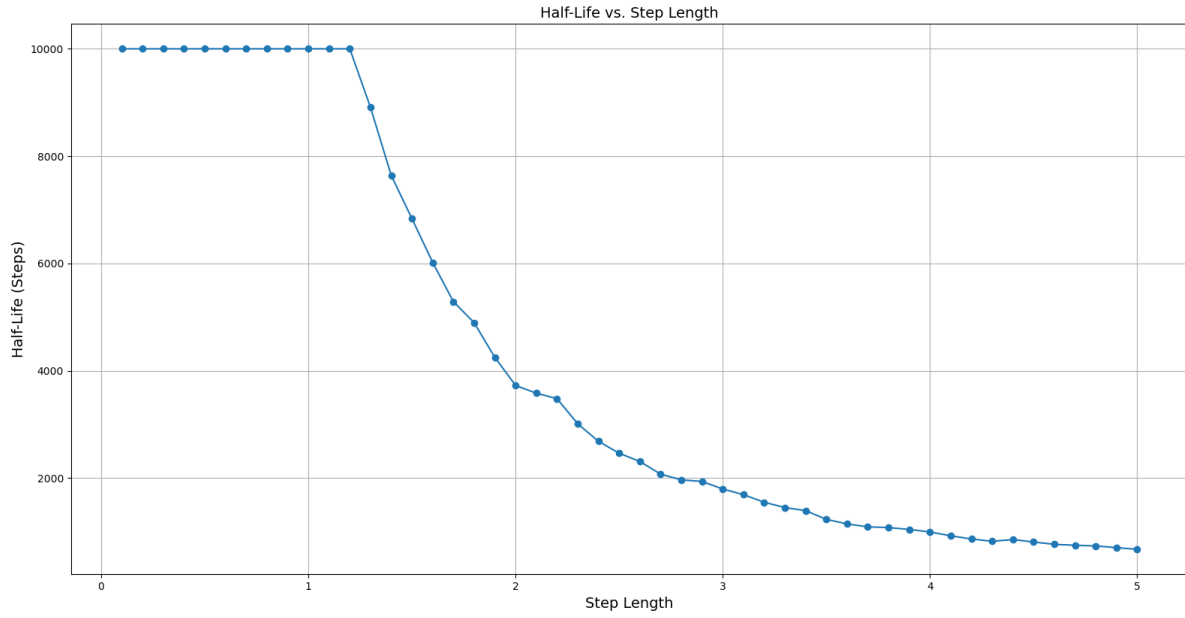


Figure 5: Half-life versus Step length

meaningful insights into diffusion, transport, and escape phenomena in constrained or bounded environments, which have broad applications in fields such as material science, biology, and fluid dynamics.

References

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- [3] H. Risken. *The Fokker-Planck Equation: Methods of Solution and Applications*. Springer, 1996.